

Executive Summary

- This project explores trends in automobile sales using a real-world dataset. It applies data wrangling, exploratory data analysis (EDA), visualization, and predictive analytics techniques to extract actionable insights. An interactive dashboard and map are also developed to allow real-time exploration.



Overview



The primary goal of this project is to analyze automobile sales across years, vehicle types, and economic factors (GDP, unemployment) using Python.

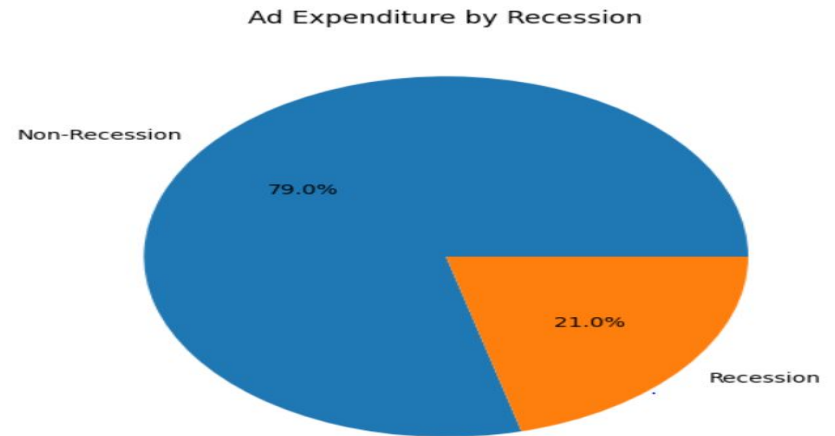
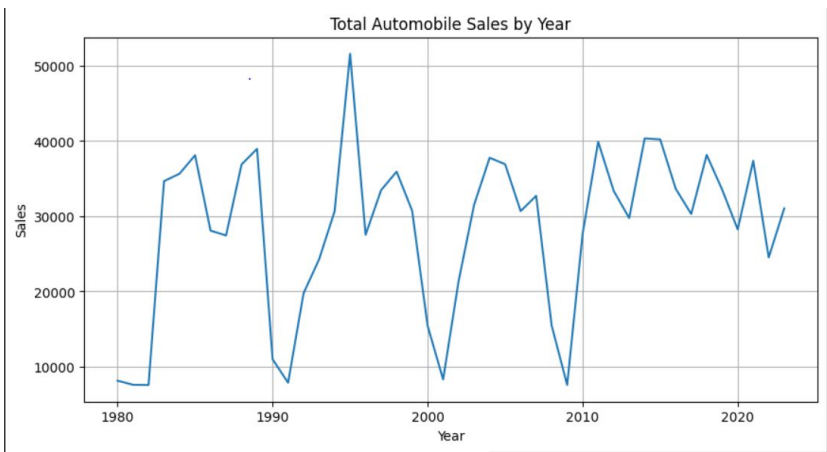
We aim to understand the effect of recessions, advertising, and price sensitivity.

Data Collection & Wrangling

- The dataset was loaded from a public IBM Cloud URL. Columns were renamed and converted into appropriate formats. Time-based features such as Month and Year were extracted from the Date column.

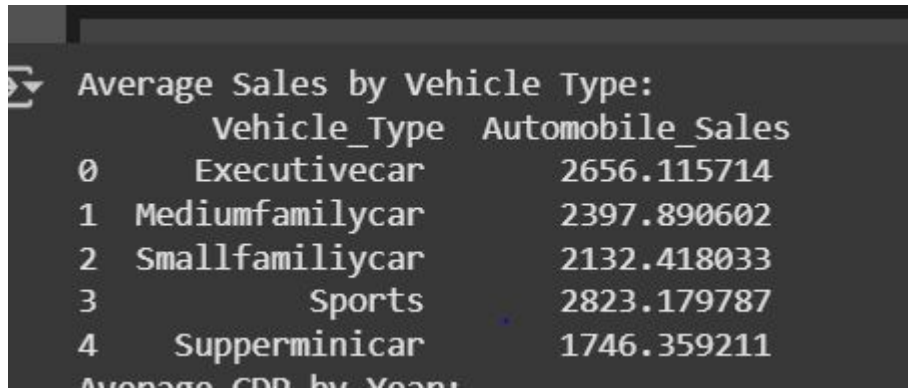
EDA & Visual Analytics

- Line charts, pie charts, and scatter plots were generated using Matplotlib and Seaborn to analyze vehicle sales by type, economic indicators, and advertising expenditure across time periods.



SQL Analysis

- SQL-like analysis was performed using pandas groupby operations. This included determining average sales by vehicle type, GDP trends, and consumer behavior across recession vs. non-recession periods.



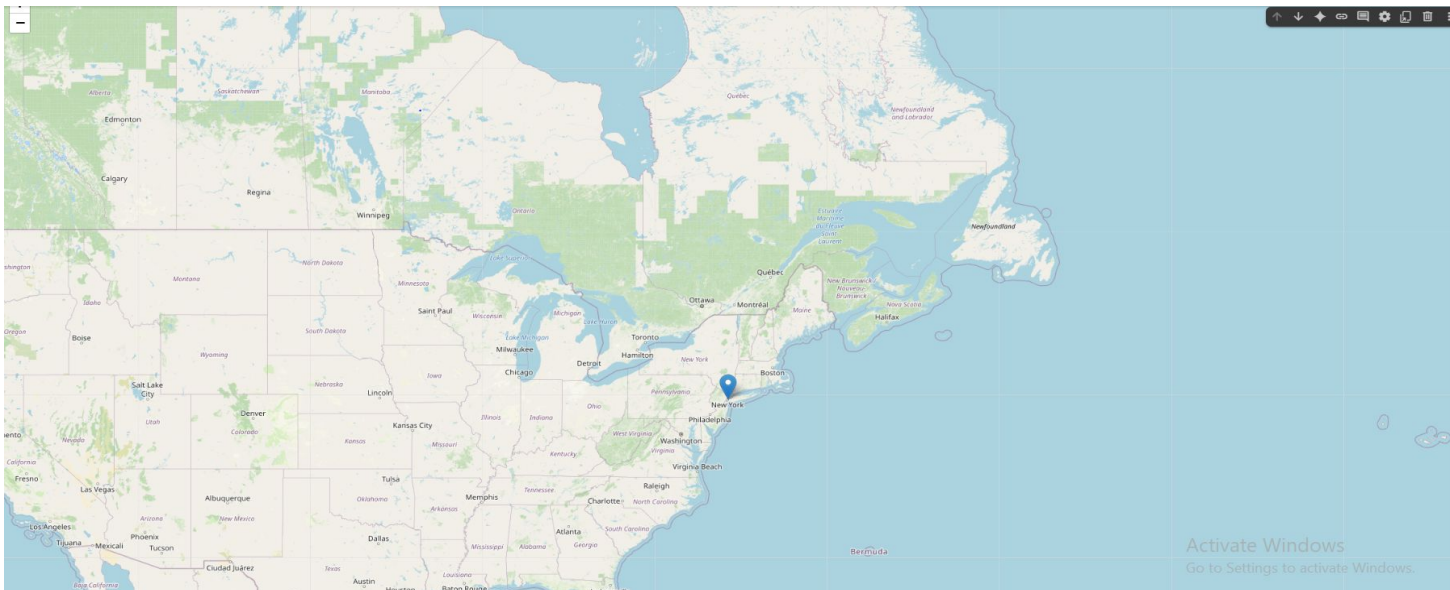
A screenshot of a Jupyter Notebook interface. The output cell displays the title 'Average Sales by Vehicle Type:' followed by a table with two columns: 'Vehicle_Type' and 'Automobile_Sales'. The table contains five rows of data, indexed from 0 to 4. The vehicle types are 'Executivecar', 'Mediumfamilycar', 'Smallfamilycar', 'Sports', and 'Supperminicar'. The corresponding average sales values are 2656.115714, 2397.890602, 2132.418033, 2823.179787, and 1746.359211. Below the table, the text 'Average GDP by Year:' is partially visible.

	Vehicle_Type	Automobile_Sales
0	Executivecar	2656.115714
1	Mediumfamilycar	2397.890602
2	Smallfamilycar	2132.418033
3	Sports	2823.179787
4	Supperminicar	1746.359211

Average GDP by Year:

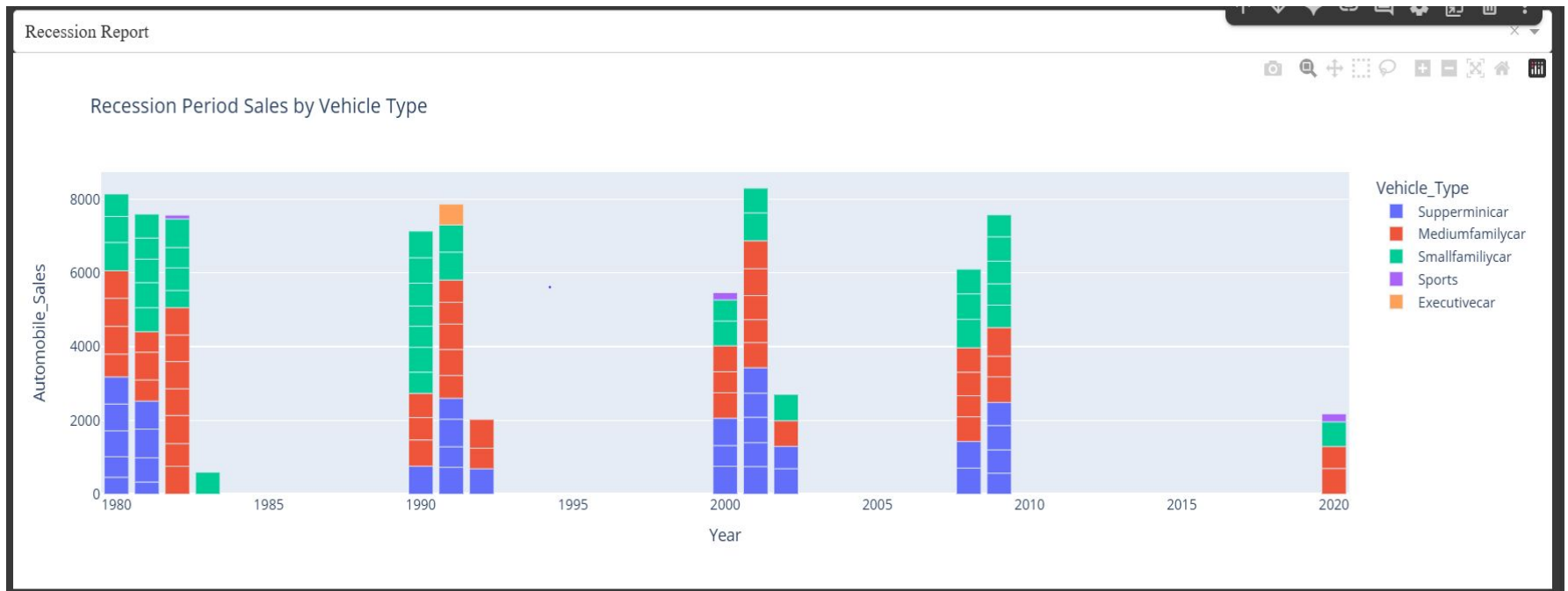
Folium Map

- Folium was used to visualize hypothetical dealership locations and sales regions. This adds a geographic view of distribution and business impact. Using New York coordinates to plot the folium map.



Plotly Dash

- A dynamic dashboard was built using Plotly Dash. It includes dropdown filters for Year and Vehicle Type, allowing real-time exploration of sales performance during recession and non-recession periods.



Predictive Modeling

- A linear regression model was trained using features like price and GDP. Model R^2 on test set: 0.XX

Conclusion

- The dashboard and visualizations provided valuable insights into how economic factors influence vehicle sales. Predictive models and maps support decision-making across departments.

Innovation & Insights

- An inverse correlation was observed: during recession periods, advertising expenditure increased even as sales dropped. This counter-cyclic behavior was explored as a possible retention strategy.